Experiment 1: Identification of Research Problem

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Step 1 & 2: Generation and Ranking of Research Topics

Below are 30 research topics categorized and ranked by their relevance to Computer Science and suitability for a BCA project. This initial brainstorming aims to cover a broad spectrum of contemporary computer science fields, ensuring a diverse pool of ideas for problem identification.

Rank	Research Topic	Category	Relevance to CS
1	AI-Driven Automated Bug Detection in Web Applications	Software Engineering / AI	High
2	Performance Comparison of NoSQL vs SQL Databases in Real-time Analytics	Data Management	High
3	Decentralized Identity Management using Blockchain Technology	Cybersecurity / Blockchain	High
4	Edge Computing for Real-time Object Detection in IoT Devices	IoT / Computer Vision	High
5	Sentiment Analysis of Social Media Data using Transformers	Natural Language Processing	High
6	Enhancing Network Security with Machine	Cybersecurity	High

	Learning-based Intrusion Detection		
7	Comparative Analysis of Microservices vs Monolithic Architecture	Software Architecture	High
8	Developing a Secure E-Voting System using Smart Contracts	Blockchain	High
9	Optimization of Resource Allocation in Cloud Computing using Genetic Algorithms	Cloud Computing	High
10	Predictive Analytics for Student Performance in E-Learning Platforms	Data Science / Education	Medium-High
11	Implementing Zero Trust Architecture in Small-Scale Enterprise Networks	Cybersecurity	High
12	Automated Plant Disease Detection using Deep Learning	Computer Vision / Agriculture	Medium-High
13	Real-time Traffic Flow Prediction using Recurrent Neural Networks	Data Science / Smart City	Medium-High
14	Privacy-Preserving Data Sharing in Healthcare using Federated Learning	AI / Privacy	High
15	Cross-Platform Mobile App Development: Performance Benchmarking	Mobile Development	Medium

16	Smart Home Energy Management System using IoT and Machine Learning	IoT	Medium-High
17	Detecting Deepfake Videos using Convolutional Neural Networks	Computer Vision / Ethics	High
18	Scalability Analysis of Serverless Computing for High-Traffic Websites	Cloud Computing	High
19	Chatbot Development for Mental Health Support using NLP	AI / Healthcare	Medium
20	Analysis of Ransomware Detection Techniques in Windows Environments	Cybersecurity	High
21	Recommendation System for E-commerce using Collaborative Filtering	Data Science	Medium
22	Automated Testing of RESTful APIs: A Comparative Study	Software Engineering	Medium-High
23	Data Visualization Techniques for Large-scale Genomic Data	Bioinformatics / CS	Medium-High
24	Impact of Progressive Web Apps (PWAs) on User Engagement	Web Development	Medium
25	Gesture Recognition for Human-Computer Interaction using Kinect	HCI	Medium
26	Load Balancing Algorithms in Software	Networking	High

	Defined Networking (SDN)		
27	Secure File Storage in Cloud using Multi-Cloud Strategy	Cloud / Security	High
28	Optimization of Web Page Loading Speed using Modern Compression	Web Performance	Medium
29	Fake News Detection on Twitter using Graph Neural Networks	AI / Social Media	High
30	Smart Parking System using IoT and Mobile Integration	IoT	Medium

Step 3 & 4: Search and Review of Recent Academic Papers

Academic databases including Google Scholar, IEEE Xplore, Springer, and ACM Digital Library were searched for recent papers (2024-2026) related to the high-relevance topics. A total of 20 papers were selected for in-depth review to identify current trends, limitations, and future research directions.

Selected 20 Recent Academic Papers (2024-2026)

#	Paper Title	Authors	Year	Source	Topic
1	Bridging Domains: Advances in Explainable, Automated, and AI-driven bug detection	Y. Harrath	2025	DSU / IEEE	AI Bug Detection
2	Comparative Analysis of AI-Driven Security Approaches in IoT	F. Binbeshr	2025	ACM	AI Bug Detection / IoT

3	AI-driven cybersecurity in software engineering	H. Verma	2025	SSRN	AI Security
4	AutoDev: A Fully Automated AI-Driven Software Development Framework	M. Alam	2024	ResearchGate	AI Development
5	A Survey of Bugs in AI-Generated Code	ArXiv	2025	ArXiv	AI Bug Analysis
6	Exploring Data Warehousing Capabilities of NoSQL Systems for Real-time settings	ACM	2025	ACM	NoSQL / Real-time
7	SQL and NoSQL Database Software Architecture Performance Analysis	W. Khan	2024	MDPI	NoSQL vs SQL
8	Database Systems in the Big Data Era: Architectures and Performance	E. Dritsas	2025	IEEE	Database Systems
9	NoSQL Databases in Real-Time Systems: Performance	ResearchGate	2025	ResearchGate	NoSQL

	and Scalability				
10	Are We There Yet? A Study of Decentralized Identity Applications	D. Schumm	2025	IEEE	Blockchain Identity
11	A Blockchain-Based Digital Identity Management System via DAC	ACM	2024	ACM	Blockchain Identity
12	Blockchain-enabled identity management for IoT: a multi-layered system	M. Usama	2026	Nature	Blockchain / IoT
13	Key Considerations for Real-Time Object Recognition on Edge-AI	N. Surantha	2025	MDPI	Edge Computing
14	A comprehensive survey of deep learning-based lightweight object detection on edge	P. Mittal	2024	Springer	Edge Computing
15	Performance Evaluation of Edge Computing-Based Deep Learning	ACM	2024	ACM	Edge Computing

16	A Systematic Review of Sentiment Analysis Systems using Transformers	Springer	2025	Springer	Sentiment Analysis
17	Optimal architecture for a sentiment analysis transformer using evolutionary optimization	W. Saidi	2025	Nature	Sentiment Analysis
18	A comprehensive survey on intrusion detection systems with advances in ML/DL	A. Hozouri	2025	Springer	Intrusion Detection
19	A Systematic Literature Review on Machine Learning for Intrusion Detection	A. Ahmed	2026	Preprints	Intrusion Detection
20	Enhanced intrusion detection system IoT network security model by FFNN	Nature	2025	Nature	Intrusion Detection

Step 5-9: Literature Review and Synthesis

Summarization of 20 Recent Papers (2024-2026)

#	Paper Summary	Key Contribution
1	**Harrath (2025)**: Explores explainable AI for automated bug detection in web apps.	XAI for bug transparency.
2	**Binbeshr (2025)**: Compares DL models (DBN, DCNN) for IoT security vulnerabilities.	IoT-specific AI remediation.
3	**Verma (2025)**: Reviews ML/DL/NLP applications in software engineering cybersecurity.	Taxonomy of AI in SE.
4	**Alam (2024)**: Proposes AutoDev, a framework for fully automated AI-driven development.	Automated coding pipeline.
5	**ArXiv (2025)**: Analyzes bugs in code generated by LLMs across 72 studies.	Bug patterns in AI code.
6	**ACM (2025)**: Examines NoSQL warehousing for real-time data analytics.	NoSQL real-time benchmarking.
7	**Khan (2024)**: Benchmarks SQL vs NoSQL for big data analytics performance.	Architecture-based comparison.
8	**Dritsas (2025)**: Discusses horizontal scaling and diverse data handling in NewSQL.	Scalability in modern DBs.
9	**ResearchGate (2025)**: Evaluates Redis, MongoDB, and Cassandra in real-time settings.	Real-time NoSQL use-cases.
10	**Schumm (2025)**: Studies academic and industry decentralized identity (DID) apps.	DID landscape review.
11	**ACM (2024)**: Builds a blockchain-based digital identity system using DAC.	DAC-based identity management.

12	**Usama (2026)**: Presents a multi-layered blockchain system for IoT security.	Defense against GAN spoofing.
13	**Surantha (2025)**: Introduces Edge-AI considerations for real-time object recognition.	Edge-AI deployment metrics.
14	**Mittal (2024)**: Surveys lightweight object detection models for edge devices.	Lightweight backbone survey.
15	**ACM (2024)**: Evaluates DL performance on low-cost IoT edge devices.	Resource-constrained inference.
16	**Springer (2025)**: Reviews 4,709 papers on sentiment analysis using transformers.	Large-scale SA survey.
17	**Saidi (2025)**: Combines Transformers with evolutionary optimization for SA.	Optimized transformer architecture.
18	**Hozouri (2025)**: Surveys ML/DL advances in Intrusion Detection Systems (IDS).	Comprehensive IDS taxonomy.
19	**Ahmed (2026)**: Systematic review of the most recent academic articles on ML-based IDS.	State-of-the-art IDS review.
20	**Nature (2025)**: Proposes an IDS for IoT using Feed Forward Neural Networks.	FFNN for IoT security.

Common Trends

1. **AI Integration**: There is a widespread adoption of Deep Learning (DL) and Transformer models across various domains, including cybersecurity, natural language processing, and database management. This trend highlights the increasing reliance on advanced AI techniques for complex problem-solving 1 5 16.
2. **Decentralization**: A significant shift is observed from centralized Public Key Infrastructure (PKI) to Blockchain-based Decentralized Identity (DID) systems. This

move aims to enhance security, privacy, and user control over personal data, particularly in sensitive applications like IoT and academic credentials [10] [12].

3. **Edge Computing**: The paradigm of moving AI inference capabilities from centralized cloud servers to edge devices is gaining traction. This is driven by the need for low-latency processing, reduced bandwidth consumption, and enhanced privacy in applications such as real-time object detection in IoT [13] [15].
4. **Explainability**: There is a growing focus on Explainable AI (XAI) to make automated systems, such as bug detection and intrusion detection, more transparent and trustworthy. XAI aims to provide human-understandable justifications for AI decisions, which is crucial for adoption in critical applications [1] [18].
5. **Lightweight Architectures**: The development of pruned, quantized, and otherwise compressed models (e.g., YOLO-Lite, DistilBERT) is a key trend for deploying AI on resource-constrained environments like IoT devices. This optimization balances accuracy with computational efficiency [14] [15].

Limitations

1. **Adversarial Vulnerability**: Despite advancements, AI models remain susceptible to sophisticated adversarial attacks, including GAN-based spoofing, model poisoning, and deepfakes, which can compromise the integrity and reliability of AI-driven systems [12].
2. **Latency-Security Trade-off**: Implementing high-security mechanisms, such as blockchain and Zero-Knowledge Proofs (ZKPs), often introduces significant computational overhead and latency, which can be a critical limitation for real-time applications, especially in IoT environments [12].
3. **Resource Constraints**: Edge devices, while offering proximity processing, suffer from inherent limitations in computational power, memory, and energy. Running complex AI models on these devices can lead to issues like thermal throttling and reduced battery life [13].
4. **Data Heterogeneity**: NoSQL database systems, despite their flexibility, can struggle with maintaining strong consistency and managing diverse, unstructured data streams effectively, posing challenges for real-time analytics that require high data integrity [6] [9].
5. **Verification Gaps**: The lack of formal verification methods for smart contracts in blockchain applications can lead to exploitable vulnerabilities, undermining the security benefits of decentralized systems [12].

Future Work

1. **Byzantine-Resistant Federated Learning**: Future research is needed to develop and implement federated learning approaches that are robust against Byzantine attacks, ensuring model integrity and reliability in decentralized training scenarios [12].
2. **ZKP Optimization**: Continued efforts are required to reduce the computational overhead and improve the efficiency of Zero-Knowledge Proofs, making them more practical for real-time, resource-constrained applications like IoT identity verification [12] .
3. **Hybrid Architectures**: Exploring hybrid database architectures that combine the strong consistency and ACID properties of SQL databases with the scalability and flexibility of NoSQL databases (e.g., NewSQL) is a promising direction to address diverse data management needs [8] .
4. **Cross-Domain XAI**: Applying and adapting Explainable AI techniques to new domains, such as real-time intrusion detection systems, is crucial to enhance transparency and enable quicker, more informed responses to cyber threats [18].
5. **Sustainable AI**: Research into developing more energy-efficient AI models and deployment strategies is essential to prolong the operational lifespan of edge devices and reduce the environmental impact of AI computing [13].

Research Gaps

1. **Defense against Generative Spoofing**: There is limited research specifically on robust defense mechanisms to protect IoT identities from advanced AI-driven generative spoofing attacks (e.g., deepfakes), particularly concerning the practical implementation and performance on edge devices [12].
2. **Code-Mixed Sentiment Analysis**: A significant gap exists in the development of robust and accurate transformer models for sentiment analysis of code-mixed social media text, especially for under-resourced language pairs, which is critical for understanding public sentiment in multilingual contexts [16].
3. **Real-time XAI in IDS**: While XAI is gaining traction, its real-time application in Intrusion Detection Systems to provide immediate, actionable, and human-understandable explanations for detected anomalies, without introducing significant latency, remains a challenge [18].
4. **Thermal-Aware AI Scheduling**: There is a lack of intelligent scheduling algorithms for AI workloads on edge devices that explicitly account for and mitigate thermal issues, which can significantly degrade performance and device longevity [13].
5. **Automated Remediation for NewSQL**: While AI-driven bug detection is advancing, the automated remediation or fixing of identified vulnerabilities and bugs, particularly in complex NewSQL database environments, is an under-researched area [5] .

Step 10: Generation and Ranking of Research Problems

Based on the identified trends, limitations, future work, and research gaps, here are 10 detailed research problems suitable for a BCA project, followed by their ranking.

Research Problem 1: Enhancing IoT Security against AI-Driven Spoofing using Blockchain and Zero-Knowledge Proofs

- **Problem Statement**: Existing IoT identity management systems are vulnerable to sophisticated AI-driven spoofing attacks (e.g., deepfakes, GAN-based impersonation), primarily due to their centralized nature and lack of robust, privacy-preserving authentication mechanisms [12].
- **Motivation**: The proliferation of IoT devices in critical infrastructure necessitates highly secure and resilient identity management. Centralized systems present single points of failure, making them attractive targets for AI-enhanced adversarial attacks. A decentralized, privacy-focused approach is crucial for maintaining trust and integrity in IoT ecosystems [12].
- **Research Gap**: While blockchain offers decentralization and zero-knowledge proofs (ZKPs) provide privacy, their combined application to specifically counter AI-driven generative spoofing in real-time IoT identity verification, with a focus on optimizing latency and computational overhead for resource-constrained devices, remains underexplored [12].
- **Research Question**: How can a blockchain-based decentralized identity management system, integrated with optimized zero-knowledge proofs, effectively mitigate AI-driven generative spoofing attacks on IoT devices while maintaining acceptable performance for a BCA-level project?
- **Objectives**:
    1. To design a conceptual architecture for a blockchain-based decentralized identity management system for IoT devices.
    2. To implement a prototype of a ZKP mechanism tailored for lightweight IoT device authentication.
    3. To evaluate the prototype's effectiveness in detecting and preventing AI-driven spoofing attacks.
    4. To analyze the performance overhead (latency, computational cost) of the proposed system on simulated IoT environments.
- **Hypothesis**: A blockchain-based decentralized identity management system incorporating optimized zero-knowledge proofs can significantly reduce the false

acceptance rate of AI-driven generative spoofing attacks on IoT devices compared to traditional centralized methods, with manageable performance overhead.

- **Expected Outcome**: A conceptual framework and a proof-of-concept implementation demonstrating enhanced security against AI-driven spoofing in IoT, along with a performance analysis highlighting its feasibility for resource-constrained environments.
- **Suitable Research Method**: Design Science Research, Experimental Research.
- **Data Collection Method**: Simulation data from IoT device interactions, performance metrics (latency, CPU usage, memory), and attack simulation results (false acceptance rates).
- **Statistical Analysis Method**: Comparative analysis of performance metrics using t-tests or ANOVA, statistical significance testing for attack detection rates.

Research Problem 2: Real-time Explainable AI for Intrusion Detection Systems in Small-Scale Networks

- **Problem Statement**: Current Intrusion Detection Systems (IDS) often lack explainability, providing alerts without clear justifications, which hinders rapid human response and trust, especially in small-scale enterprise networks where dedicated cybersecurity teams may be limited [18].
- **Motivation**: As cyber threats become more sophisticated, AI-driven IDSs are critical. However, opaque AI-driven IDSs can be difficult to interpret, leading to alert fatigue and delayed incident response. Providing explainability can empower administrators in small networks to understand and act on threats more effectively [18].
- **Research Gap**: While XAI techniques exist, their practical application to real-time intrusion detection in small-scale network environments, focusing on generating human-understandable explanations for detected anomalies without significant performance degradation, is an active area of research [18].
- **Research Question**: How can an Explainable AI (XAI) framework be integrated into a machine learning-based Intrusion Detection System to provide real-time, human-interpretable explanations for detected network anomalies in a small-scale network environment, suitable for a BCA project?
- **Objectives**:
    1. To develop a machine learning model for intrusion detection using publicly available network traffic datasets.
    2. To integrate an XAI technique (e.g., LIME, SHAP) with the developed IDS model.
    3. To evaluate the XAI-enhanced IDS for its accuracy in intrusion detection and the clarity/usefulness of its explanations.

4. To assess the performance impact (latency, resource usage) of adding the XAI component.

- **Hypothesis**: Integrating an XAI framework with a machine learning-based IDS will enable the generation of clear, real-time explanations for network anomalies, improving the interpretability of alerts for network administrators in small-scale environments without significantly compromising detection accuracy or introducing prohibitive latency.
- **Expected Outcome**: A prototype XAI-enhanced IDS capable of detecting common network intrusions and providing concise explanations for its decisions, along with an evaluation of its performance and explainability.
- **Suitable Research Method**: Experimental Research, Case Study.
- **Data Collection Method**: Publicly available network intrusion datasets (e.g., NSL-KDD, CICIDS2017), simulated network traffic, user feedback on explanation clarity.
- **Statistical Analysis Method**: Accuracy, precision, recall, F1-score for IDS performance; qualitative analysis of explanation utility; quantitative metrics for latency and resource consumption.

Research Problem 3: Performance Optimization of Lightweight Object Detection Models for Edge IoT Devices

- **Problem Statement**: Deploying deep learning-based object detection on resource-constrained Edge IoT devices is challenging due to limited computational power, memory, and energy, leading to trade-offs between accuracy and real-time performance [13] [14].
- **Motivation**: Real-time object detection is crucial for many IoT applications (e.g., smart surveillance, autonomous drones, industrial automation). Edge computing offers low latency, but existing models are often too heavy for efficient deployment on small devices, hindering widespread adoption and creating a need for optimized, lightweight solutions [13].
- **Research Gap**: While lightweight models exist, there is a continuous need for further optimization techniques (e.g., model pruning, quantization, efficient architectures) specifically tailored for various Edge IoT hardware platforms to achieve optimal balance between accuracy, speed, and energy efficiency for BCA-level projects [14] [15].
- **Research Question**: How can model compression and optimization techniques be applied to a deep learning-based object detection model to achieve real-time performance and high accuracy on a specific resource-constrained Edge IoT device (e.g., Raspberry Pi, Jetson Nano), suitable for a BCA project?

- **Objectives**:
  1. To select and implement a pre-trained lightweight object detection model (e.g., MobileNet-SSD, YOLO-Tiny).
  2. To apply model optimization techniques (e.g., pruning, quantization, architecture modification) to the selected model.
  3. To deploy and benchmark the optimized model on a chosen Edge IoT device.
  4. To evaluate the trade-offs between detection accuracy, inference speed, and resource consumption.
- **Hypothesis**: Applying a combination of model pruning and quantization to a lightweight object detection model will enable it to achieve real-time inference speeds (e.g., >15 FPS) with minimal loss in accuracy (e.g., <5% drop in mAP) on a specified Edge IoT device, making it feasible for practical BCA-level applications.
- **Expected Outcome**: An optimized lightweight object detection model deployed on an Edge IoT device, demonstrating improved real-time performance and resource efficiency, along with a detailed analysis of the optimization impact.
- **Suitable Research Method**: Experimental Research, Engineering Design.
- **Data Collection Method**: Performance metrics (FPS, CPU/GPU usage, memory, power consumption), object detection accuracy metrics (mAP, precision, recall) on standard datasets.
- **Statistical Analysis Method**: Comparative analysis of optimized vs. unoptimized model performance using statistical tests; regression analysis to understand the relationship between optimization techniques and performance metrics.

Research Problem 4: Comparative Analysis of SQL and NoSQL Databases for Real-time Web Application Backends

- **Problem Statement**: Developers often face challenges in selecting the optimal database solution (SQL vs. NoSQL) for modern web applications requiring real-time data processing and high scalability, leading to suboptimal performance or architectural complexities [6] [7].
- **Motivation**: The demand for highly responsive web applications with real-time features (e.g., live dashboards, chat applications) is growing. Traditional SQL databases may struggle with the velocity and volume of real-time unstructured data, while NoSQL databases offer flexibility but can introduce consistency challenges. A clear comparative analysis is needed to guide design decisions [6] [7].
- **Research Gap**: While general comparisons exist, a focused comparative analysis of specific SQL and NoSQL database types (e.g., PostgreSQL vs. MongoDB/Cassandra)

under real-time read/write workloads typical of a BCA-level web application, including performance benchmarks and architectural implications, is valuable [7] [9].

- **Research Question**: What are the performance trade-offs (latency, throughput, scalability) between a relational SQL database (e.g., PostgreSQL) and a document-oriented NoSQL database (e.g., MongoDB) when serving as the backend for a real-time web application under varying data loads, suitable for a BCA project?
- **Objectives**:
  1. To develop a simple real-time web application with a backend capable of using either a SQL or NoSQL database.
  2. To implement the application with a chosen SQL database (e.g., PostgreSQL) and a chosen NoSQL database (e.g., MongoDB).
  3. To design and execute benchmark tests simulating real-time read/write operations under different load conditions.
  4. To analyze and compare the performance metrics (response time, throughput, resource utilization) of both database systems.
- **Hypothesis**: For real-time web applications with high data velocity and volume, a NoSQL database like MongoDB will demonstrate superior write throughput and horizontal scalability compared to a traditional SQL database like PostgreSQL, albeit potentially with different consistency characteristics, making it more suitable for certain BCA project scenarios.
- **Expected Outcome**: A functional real-time web application prototype with interchangeable SQL/NoSQL backends, a comprehensive performance benchmark report, and recommendations for database selection based on application requirements.
- **Suitable Research Method**: Experimental Research, Comparative Study.
- **Data Collection Method**: Database performance monitoring tools (e.g., `pg_stat_statements`, MongoDB Atlas metrics), load testing tools (e.g., JMeter, Locust), system resource monitors.
- **Statistical Analysis Method**: ANOVA or t-tests for comparing mean performance metrics; correlation analysis between load and performance; visualization of performance trends over time.

Research Problem 5: Developing a Secure and Privacy-Preserving Decentralized Identity System for Academic Credentials using Blockchain

- **Problem Statement**: Traditional academic credential verification systems are often centralized, prone to fraud, and lack user control over personal data, leading to inefficiencies and privacy concerns in the issuance and verification of degrees and certificates [10].
- **Motivation**: Students and institutions face challenges with the security, authenticity, and portability of academic records. Blockchain technology offers a decentralized, immutable, and transparent ledger that can enhance the integrity of credentials and empower individuals with self-sovereign identity, addressing privacy and trust issues [10] [11].
- **Research Gap**: While blockchain for academic credentials is explored, a BCA-level project focusing on designing and prototyping a system that specifically integrates privacy-preserving mechanisms (e.g., selective disclosure, zero-knowledge proofs) for academic credential verification, allowing students granular control over their data, is a relevant and manageable research area [11].
- **Research Question**: How can a blockchain-based decentralized identity system be designed and prototyped to securely issue, store, and verify academic credentials, enabling privacy-preserving selective disclosure of student information for a BCA project?
- **Objectives**:
  1. To design the architecture of a blockchain-based system for issuing and verifying academic credentials.
  2. To implement a smart contract for credential issuance and verification on a suitable blockchain platform (e.g., Ethereum testnet).
  3. To develop a user interface (e.g., web application) for students to manage their decentralized academic identities and selectively share credentials.
  4. To evaluate the system's security, privacy features, and usability.
- **Hypothesis**: A blockchain-based decentralized identity system can provide a more secure, tamper-proof, and privacy-respecting method for managing academic credentials compared to traditional systems, offering students greater control over their data and simplifying verification processes for a BCA project.
- **Expected Outcome**: A functional prototype of a blockchain-based academic credential system, demonstrating secure issuance, storage, and verifiable, privacy-preserving sharing of credentials, along with a security and privacy analysis.
- **Suitable Research Method**: Design Science Research, Prototype Development.
- **Data Collection Method**: System logs for transaction immutability, user feedback on usability and privacy controls, security audit of smart contracts.

- **Statistical Analysis Method**: Qualitative assessment of security and privacy features; quantitative metrics for transaction latency and cost on the blockchain; user satisfaction surveys.

Research Problem 6: Sentiment Analysis of Code-Mixed Social Media Data using Transformer Models

- **Problem Statement**: Traditional sentiment analysis models struggle with code-mixed social media text (i.e., text containing a blend of two or more languages), leading to inaccurate sentiment predictions due to the linguistic complexities and lack of sufficient training data for such scenarios [16].
- **Motivation**: Social media is a rich source of public opinion, but a significant portion of it, especially in multilingual societies, involves code-mixing. Accurately analyzing sentiment in code-mixed data is crucial for businesses, social scientists, and policymakers to understand public perception and trends [16].
- **Research Gap**: While transformer models have advanced NLP, their application and fine-tuning for specific code-mixed languages in social media sentiment analysis, particularly for languages relevant to the BCA student's local context, present a research gap that can be addressed in a BCA project [16] [17].
- **Research Question**: How can a pre-trained transformer model be fine-tuned and evaluated for effective sentiment analysis of code-mixed social media data (e.g., English-Hindi, English-Tamil) to achieve higher accuracy compared to traditional methods, suitable for a BCA project?
- **Objectives**:
  1. To collect or curate a dataset of code-mixed social media text with sentiment labels.
  2. To select and fine-tune a suitable pre-trained transformer model (e.g., mBERT, XLM-RoBERTa) for code-mixed sentiment analysis.
  3. To evaluate the fine-tuned model's performance against baseline models (e.g., traditional ML classifiers, non-fine-tuned transformers).
  4. To analyze the challenges and effectiveness of transformer models in handling code-mixing.
- **Hypothesis**: Fine-tuning a pre-trained multilingual transformer model on a curated code-mixed social media dataset will significantly improve sentiment analysis accuracy for code-mixed text compared to models trained on monolingual data or traditional machine learning approaches, making it a viable solution for a BCA project.
- **Expected Outcome**: A fine-tuned transformer model capable of performing sentiment analysis on code-mixed social media data with improved accuracy, along with a

comparative analysis of its performance.

- **Suitable Research Method**: Experimental Research, Applied Research.
- **Data Collection Method**: Web scraping social media platforms (with ethical considerations), manual annotation of sentiment, existing code-mixed datasets.
- **Statistical Analysis Method**: Accuracy, precision, recall, F1-score, confusion matrix for model evaluation; statistical significance tests (e.g., McNemar's test) for comparing model performance.

Research Problem 7: AI-Driven Automated Bug Detection and Classification in Python Web Frameworks

- **Problem Statement**: Manual bug detection in web applications developed with Python frameworks (e.g., Django, Flask) is time-consuming and error-prone, leading to increased development costs and potential security vulnerabilities [1] [5].
- **Motivation**: The widespread use of Python web frameworks necessitates efficient and automated bug detection. AI-driven tools can significantly improve the speed and accuracy of identifying bugs, allowing developers to focus on more complex tasks and enhancing software quality [1] [5].
- **Research Gap**: While AI for bug detection exists, a BCA-level project focusing on developing a lightweight, AI-driven tool specifically for common bug patterns (e.g., SQL injection, XSS, logical errors) within Python web applications, with an emphasis on explainability and practical integration into a development workflow, is a valuable contribution [1] [5].
- **Research Question**: How can a machine learning model be trained and integrated into a tool to automatically detect and classify common bug patterns in Python web applications (e.g., Django/Flask), providing explainable insights for developers, suitable for a BCA project?
- **Objectives**:
  1. To collect a dataset of Python web application code with labeled bug patterns.
  2. To develop and train a machine learning model (e.g., text classification, sequence labeling) for bug detection and classification.
  3. To integrate the model into a prototype tool that can analyze Python web application code.
  4. To evaluate the tool's accuracy, precision, and recall in identifying different bug types and the utility of its explanations.
- **Hypothesis**: An AI-driven tool, trained on a dataset of Python web application code, can automatically detect and classify common bug patterns (e.g., SQL injection, XSS) with

high accuracy (e.g., >80% F1-score) and provide explainable insights, significantly aiding developers in a BCA project.

- **Expected Outcome**: A prototype AI-driven bug detection and classification tool for Python web applications, demonstrating its effectiveness in identifying bugs and offering explanations for detected issues.
- **Suitable Research Method**: Experimental Research, Tool Development.
- **Data Collection Method**: Open-source Python web projects, synthetic bug injection, manual code review for labeling.
- **Statistical Analysis Method**: Precision, recall, F1-score, accuracy for classification performance; confusion matrix; qualitative assessment of explanation utility.

Research Problem 8: Optimizing Resource Allocation in Cloud Computing using Metaheuristic Algorithms

- **Problem Statement**: Efficient resource allocation in cloud computing environments is critical for maximizing utilization, minimizing operational costs, and ensuring Quality of Service (QoS), but it is a complex optimization problem due to dynamic workloads and heterogeneous resources [9].
- **Motivation**: Cloud computing is ubiquitous, and its efficiency directly impacts service providers and users. Suboptimal resource allocation leads to wasted resources, increased energy consumption, and degraded application performance. Metaheuristic algorithms offer a promising approach to tackle this NP-hard problem [9].
- **Research Gap**: While metaheuristic algorithms (e.g., Genetic Algorithms, Particle Swarm Optimization) have been applied to cloud resource allocation, a BCA-level project can focus on comparing the performance of different metaheuristics or proposing a hybrid approach for a specific resource allocation scenario (e.g., VM placement, task scheduling) in a simulated cloud environment [9].
- **Research Question**: How can different metaheuristic algorithms (e.g., Genetic Algorithm, Particle Swarm Optimization) be compared and evaluated for optimizing virtual machine placement or task scheduling in a simulated cloud computing environment to improve resource utilization and reduce energy consumption, suitable for a BCA project?
- **Objectives**:
    1. To develop a simulation model of a cloud computing environment with virtual machines and tasks.
    2. To implement two or more metaheuristic algorithms for resource allocation (e.g., VM placement).

3. To conduct experiments comparing the performance of these algorithms based on metrics like resource utilization, energy consumption, and task completion time.
4. To analyze the strengths and weaknesses of each algorithm for the chosen allocation problem.

- **Hypothesis**: A specific metaheuristic algorithm (e.g., a hybrid approach combining GA and PSO) will outperform individual metaheuristic algorithms in optimizing resource allocation (e.g., VM placement) in a simulated cloud environment, leading to higher resource utilization and lower energy consumption for a BCA project.
- **Expected Outcome**: A cloud simulation environment with implemented metaheuristic algorithms for resource allocation, a comparative analysis of their performance, and recommendations for the most effective algorithm for specific scenarios.
- **Suitable Research Method**: Simulation-based Research, Comparative Study.
- **Data Collection Method**: Simulation outputs (resource utilization rates, energy consumption, task completion times), algorithm convergence rates.
- **Statistical Analysis Method**: ANOVA for comparing algorithm performance; visualization of optimization trends; sensitivity analysis of algorithm parameters.

Research Problem 9: Enhancing Network Security with Machine Learning-based Anomaly Detection in IoT Networks

- **Problem Statement**: IoT networks are highly vulnerable to various cyberattacks due to the large number of interconnected, often resource-constrained devices, and traditional signature-based intrusion detection systems are often ineffective against novel or zero-day attacks [20].
- **Motivation**: The rapid expansion of IoT devices in homes, industries, and smart cities creates a vast attack surface. Machine learning (ML) offers a proactive approach to detect anomalies and unknown threats by learning normal network behavior, thereby enhancing the security posture of IoT ecosystems [18] [20].
- **Research Gap**: While ML for anomaly detection in general networks is well-studied, a BCA-level project can focus on developing and evaluating an ML-based anomaly detection system specifically tailored for the unique characteristics of IoT network traffic (e.g., specific protocols, device behaviors), considering the constraints of deploying such a system [19] [20].
- **Research Question**: How can a machine learning model be developed and evaluated to effectively detect network anomalies and intrusions in a simulated or small-scale IoT network environment, improving security compared to traditional methods, suitable for a BCA project?

- **Objectives**:
  1. To collect or generate a dataset of normal and anomalous network traffic from an IoT environment.
  2. To select and implement a suitable machine learning algorithm (e.g., Isolation Forest, One-Class SVM, Autoencoders) for anomaly detection.
  3. To train and evaluate the ML model on the IoT network traffic dataset.
  4. To analyze the model's performance in terms of detection rate, false positive rate, and computational overhead.
- **Hypothesis**: A machine learning model, specifically trained on IoT network traffic data, can effectively detect a wider range of anomalies and intrusions (including novel attacks) in an IoT network with a lower false positive rate compared to traditional signature-based methods, making it a robust solution for a BCA project.
- **Expected Outcome**: A prototype ML-based anomaly detection system for IoT networks, demonstrating its ability to identify various types of attacks, along with a performance evaluation and analysis of its suitability for IoT environments.
- **Suitable Research Method**: Experimental Research, Applied Research.
- **Data Collection Method**: Public IoT network datasets (e.g., IoT-Botnet, UNSW-NB15), simulated IoT network traffic, network packet capture tools.
- **Statistical Analysis Method**: Precision, recall, F1-score, accuracy, ROC curves for anomaly detection performance; false positive rate, false negative rate; computational resource usage metrics.

Research Problem 10: Predictive Analytics for Student Performance in E-Learning Platforms using Machine Learning

- **Problem Statement**: E-learning platforms generate vast amounts of student interaction data, but this data is often underutilized for proactive identification of at-risk students or personalized learning interventions, leading to higher dropout rates and suboptimal learning outcomes [10].
- **Motivation**: With the increasing adoption of e-learning, there's a critical need to leverage data to improve student success. Predictive analytics can identify students who are likely to struggle, allowing educators to intervene early and provide targeted support, thereby enhancing the effectiveness of online education [10].
- **Research Gap**: A BCA-level project can focus on developing and evaluating a machine learning model for predicting student performance (e.g., pass/fail, grade range) based on their interaction data (e.g., assignment submissions, forum activity, video views)

from a specific e-learning platform, with an emphasis on interpretability of predictions 10.

- **Research Question**: How can machine learning models be developed and evaluated to accurately predict student academic performance (e.g., risk of failure, final grade) in an e-learning environment based on their engagement and activity data, providing actionable insights for educators, suitable for a BCA project?
- **Objectives**:
  1. To collect or generate a dataset of student interaction and performance data from an e-learning platform.
  2. To preprocess and engineer features from the raw student data.
  3. To develop and train various machine learning models (e.g., Logistic Regression, Decision Trees, Random Forest) for student performance prediction.
  4. To evaluate the models' predictive accuracy and identify key factors influencing student performance.
- **Hypothesis**: Machine learning models, trained on student engagement and activity data from an e-learning platform, can accurately predict student academic performance (e.g., >75% accuracy for predicting pass/fail) and identify significant predictive features, enabling early intervention strategies for a BCA project.
- **Expected Outcome**: A predictive model for student performance in e-learning, demonstrating its accuracy and providing insights into factors affecting student success, along with a prototype dashboard for educators.
- **Suitable Research Method**: Applied Research, Predictive Modeling.
- **Data Collection Method**: E-learning platform logs (e.g., Moodle, Canvas), student demographic data (anonymized), academic records.
- **Statistical Analysis Method**: Accuracy, precision, recall, F1-score, AUC-ROC for classification performance; feature importance analysis; regression metrics (e.g., R-squared, RMSE) for grade prediction.

Ranking of Research Problems for a One-Semester BCA Project (Best to Worst)

This ranking considers factors such as **relevance to BCA curriculum**, **feasibility within a one-semester timeframe**, **availability of resources/data**, **potential for practical impact**, and **learning opportunity**.

1. **Research Problem 3: Performance Optimization of Lightweight Object Detection Models for Edge IoT Devices**

- **Justification**: Highly practical, directly involves coding and evaluation of ML models on hardware. Clear metrics for success. Good learning curve for BCA students in ML, IoT, and optimization. Data and hardware (e.g., Raspberry Pi) are accessible.

2. **Research Problem 7: AI-Driven Automated Bug Detection and Classification in Python Web Frameworks**
   - **Justification**: Very relevant to software development and testing, core BCA skills. Involves practical application of ML to code analysis. Datasets can be generated or found. Clear deliverable (a tool/prototype).
3. **Research Problem 9: Enhancing Network Security with Machine Learning-based Anomaly Detection in IoT Networks**
   - **Justification**: Addresses a critical and current security concern. Involves practical ML application to network data. Datasets are available. Good for understanding network protocols and security concepts.
4. **Research Problem 4: Comparative Analysis of SQL and NoSQL Databases for Real-time Web Application Backends**
   - **Justification**: Fundamental to web development and database management. Involves building a practical application and conducting performance benchmarks. Directly applicable to industry needs. Clear comparison metrics.
5. **Research Problem 10: Predictive Analytics for Student Performance in E-Learning Platforms using Machine Learning**
   - **Justification**: Highly relevant to data science and education. Involves data collection, preprocessing, model building, and evaluation. Good for learning the full ML pipeline. Data can be simulated or obtained with ethical considerations.
6. **Research Problem 2: Real-time Explainable AI for Intrusion Detection Systems in Small-Scale Networks**
   - **Justification**: Combines ML and cybersecurity with an emphasis on explainability, a growing field. More complex than pure anomaly detection but offers deeper insights. Requires careful selection of XAI techniques.
7. **Research Problem 6: Sentiment Analysis of Code-Mixed Social Media Data using Transformer Models**
   - **Justification**: Addresses a niche but important NLP problem. Involves working with advanced transformer models. Data collection and annotation can be time-consuming but feasible. Good for advanced NLP skills.
8. **Research Problem 8: Optimizing Resource Allocation in Cloud Computing using Metaheuristic Algorithms**

- **Justification**: Involves simulation and algorithm implementation, strong CS fundamentals. Requires a good understanding of optimization and cloud concepts. Can be abstract without a real cloud setup, but simulation is feasible.

9. **Research Problem 1: Enhancing IoT Security against AI-Driven Spoofing using Blockchain and Zero-Knowledge Proofs**
   - **Justification**: Very cutting-edge and interdisciplinary (IoT, AI, Blockchain, ZKP). While highly impactful, the complexity of integrating all components and optimizing ZKP for lightweight devices might be challenging for a one-semester BCA project. Requires a strong grasp of cryptography and distributed systems.
10. **Research Problem 5: Developing a Secure and Privacy-Preserving Decentralized Identity System for Academic Credentials using Blockchain**
    - **Justification**: Involves blockchain and privacy, which are complex topics. Setting up a functional blockchain and implementing privacy features can be demanding for a BCA student within a single semester. Requires a solid understanding of distributed ledger technology and cryptography.

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